

Exercise Sheet 7

1.

$H_\ell(x)$, $\ell \in \mathbb{Z}_{\geq 0}$ defined via

$$H_\ell(x) := \frac{\ell!}{2\pi i} \int_{\Gamma} e^{2xz - z^2} \frac{dz}{z^{\ell+1}}$$

unit circle clockwise

$$\Gamma = \{z \in \mathbb{C} : |z|=1\} \curvearrowright$$

(a) First find H_0 :

$$H_0(x) = \frac{0!}{2\pi i} \oint_{\Gamma} \frac{e^{2xz - z^2}}{z} dz$$

Let $f(z) = \frac{e^{2xz - z^2}}{z}$. By residue theorem

$$\oint_{\Gamma} f(z) dz = 2\pi i \operatorname{Res} f(z)$$

Simple pole at $z=0$, so have $\operatorname{Res} f(z) = \lim_{z \rightarrow 0} f(z) = \lim_{z \rightarrow 0} \frac{e^{2xz - z^2}}{z} = 1$. Hence

$$\underline{H_0(x) = \frac{1}{2\pi i} \cdot 2\pi i = 1}$$

For H_1 , take similar approach.

$$H_1 = \frac{1!}{2\pi i} \oint_{\Gamma} f(z) dz$$

where $f(z) = \frac{e^{2xz - z^2}}{z^2}$. Two poles (still $z=0$). Can use

$$\operatorname{Res} f = \lim_{z \rightarrow 0} \frac{d}{dz} [z^2 f(z)]$$

$$= \lim_{z \rightarrow 0} (2x - 2z) e^{2xz - z^2} = 2x$$

Hence

$$\underline{H_1(x) = \frac{1}{2\pi i} \cdot 2\pi i \cdot 2x = 2x}$$

For H_2 take similar approach again.

$$H_2(x) = \frac{2!}{2\pi i} \oint_{\Gamma} f(z) dz$$

where $f(z) = \frac{e^{2xz - z^2}}{z^3}$. Three poles ($z=0$), so

$$\text{Res } f = \frac{1}{2!} \lim_{z \rightarrow 0} \frac{d^2}{dz^2} [z^3 f(z)]$$

$$= \frac{1}{2} \lim_{z \rightarrow 0} \frac{d}{dz} (2x - 2z) e^{2xz - z^2}$$

$$= \frac{1}{2} \lim_{z \rightarrow 0} \left[2x(2x - 2z) e^{2xz - z^2} - 2e^{2xz - z^2} - 2z(2x - 2z) e^{2xz - z^2} \right]$$

$$= 2x^2 - 1$$

Hence

$$\underline{H_2(x) = \frac{2}{2\pi i} \cdot 2\pi i \cdot (2x^2 - 1) = 4x^2 - 2}$$

(b) To be orthogonal w.r.t. a weight $w(x) = e^{-x^2}$ they must all satisfy

$$\int_{-\infty}^{\infty} H_i(x) H_j(x) w(x) dx = N \delta_{ij}$$

So, must check $(i, j) = (0, 0), (0, 1), (0, 2), (1, 1), (1, 2), (2, 2)$. The general case (up to order x^2) with

$$H_i(x) = ax^2 + bx + c, \quad H_j(x) = dx^2 + fx + g$$

s.t.

$$\int_{-\infty}^{\infty} H_i(x) H_j(x) w(x) dx = \int_{-\infty}^{\infty} (ax^2 + bx + c)(dx^2 + fx + g) e^{-x^2} dx$$

$$= \int_{-\infty}^{\infty} [adx^4 + (af + bd)x^3 + (ag + bf + cd)x^2 + (bg + cf)x + cg] e^{-x^2} dx$$

$$= \sum_{n=0}^4 a_n \int_{-\infty}^{\infty} x^n e^{-x^2} dx$$

With a_n being coefficients in the second line $a_0 = cg$, $a_1 = bg + cf$, ...

The integral has a general result:

$$I_n = \int_{-\infty}^{\infty} x^n e^{-x^2} dx = \begin{cases} 0 & n \text{ odd} \\ \frac{\sqrt{\pi}(n-1)!!}{2^{n/2}} & n \text{ even} \end{cases}$$

Hence

$$\int_{-\infty}^{\infty} H_i(x) H_j(x) w(x) dx = \sqrt{\pi} \sum_{m=0}^2 a_{2m} \frac{(2m-1)!!}{2^m}$$

$$= \sqrt{\pi} \left(cg + \frac{1}{2}[ag + bf + cd] + \frac{3}{4}ad \right)$$

i	j	$H_i(x)$	$H_j(x)$	cg	ag	bf	cd	ad	$cg + \frac{1}{2}[ag + bf + cd] + \frac{3}{4}ad$
0	0	$c=1$	$g=1$	1	0	0	0	0	<u>1</u>
0	1	$c=1$	$g=2$	0	0	0	0	0	<u>0</u>
0	2	$c=1$	$d=4$ $g=-2$	-2	0	0	4	0	$-2 + \frac{4}{2} = 0$
1	1	$b=2$	$f=2$	0	0	2	0	0	<u>2</u>
1	2	$b=2$	$d=4$ $g=-2$	0	0	0	0	0	<u>0</u>
2	2	$a=4$ $c=-2$	$d=4$ $g=-2$	4	-8	0	-8	16	$4 + \frac{1}{2}[-16] + \frac{3}{4}16 = 8$

Hence when $i \neq j$, terms vanish, and when $i=j$ terms exist, these Hermite polynomials are orthogonal.

$$\int_{-\infty}^{\infty} H_i(x) H_j(x) w(x) dx = k \sqrt{\pi} \delta_{ij}$$

where k depends on which polynomial we consider.

From notes
 $k = 2^i i!$
 $= 1, 2, 8, \dots$
 as we see above

(c) For the rescaled GUE we have

$$K_N(x, y) = \sqrt{w(x)w(y)} \sum_{j=0}^{N-1} \frac{H_j(x) H_j(y)}{(H_j, H_j)}$$

$$\therefore K_3(x, y) = e^{-\frac{x^2+y^2}{2}} \left(\frac{H_0(x)H_0(y)}{\sqrt{\pi}} + \frac{H_1(x)H_1(y)}{2\sqrt{\pi}} + \frac{H_2(x)H_2(y)}{8\sqrt{\pi}} \right)$$

$$= \frac{1}{\sqrt{\pi}} e^{-\frac{x^2+y^2}{2}} \left(1 + \frac{1}{2} xy + \frac{1}{8} (4x^2-2)(4y^2-2) \right)$$

$$= \frac{1}{\sqrt{\pi}} e^{-\frac{x^2+y^2}{2}} \left(1 + \frac{1}{2} xy + 2x^2y^2 - x^2 - y^2 + \frac{1}{2} \right)$$

$$\therefore \underline{k_3(x,y) = \frac{1}{\sqrt{\pi}} e^{-\frac{x^2+y^2}{2}} \left(\frac{1}{2} - x^2 - y^2 + \frac{1}{2} xy + 2x^2y^2 \right)}$$

2.

We have

$$\frac{d^2}{dx^2} H_\ell(x) - 2x \frac{d}{dx} H_\ell(x) + 2\ell H_\ell(x) = 0$$

We want to show that

$$\psi_\ell^{(N)}(x) = e^{-\frac{N}{2}x^2} N^{1/4} \gamma_\ell H_\ell(\sqrt{N}x)$$

satisfies

$$\frac{d^2}{dx^2} \psi_\ell^{(N)}(x) = N(Nx^2 - (2\ell+1)) \psi_\ell^{(N)}(x) \quad \left(\gamma_j = \sqrt{\frac{1}{\sqrt{\pi} 2^j j!}} \right)$$

Rearrange $\psi_\ell^{(N)}(x) = e^{-Nx^2/2} N^{1/4} \gamma_\ell H_\ell(\sqrt{N}x)$ in terms of $H_\ell(\sqrt{N}x)$

$$H_\ell(\sqrt{N}x) = e^{Nx^2/2} N^{-1/4} \frac{1}{\gamma_\ell} \psi_\ell^{(N)}(x)$$

then plug into first ODE:

$$\frac{d^2}{dy^2} H_\ell(y) - 2y \frac{d}{dy} H_\ell(y) + 2\ell H_\ell(y) = 0 \quad \text{with } y = \sqrt{N}x$$

$$\text{Then, } H_\ell(y) = e^{y^2/2} N^{-1/4} \frac{1}{\gamma_\ell} \psi_\ell^{(N)}\left(\frac{y}{\sqrt{N}}\right)$$

Evaluate derivatives (set $C = N^{-1/4} \frac{1}{\gamma_\ell}$)

$$\frac{d}{dy} H_\ell(y) = C \frac{d}{dy} \left[e^{y^2/2} \cdot \psi_\ell^{(N)}\left(\frac{y}{\sqrt{N}}\right) \right]$$

$$= C \left[y e^{y^2/2} \psi_\ell^{(N)}\left(\frac{y}{\sqrt{N}}\right) + e^{y^2/2} \frac{d}{dy} \psi_\ell^{(N)}\left(\frac{y}{\sqrt{N}}\right) \right]$$

can use chain rule

$$\frac{d}{dy} \psi\left(\frac{y}{\sqrt{N}}\right) = \frac{d}{dx} \psi(x) \frac{dx}{dy} = \frac{d}{dx} \psi(x) \cdot \frac{1}{\sqrt{N}}$$

$$\frac{d}{dx} e^{ax^2} = 2x a e^{ax^2}$$

$$\begin{aligned}
&= c e^{y^2/2} \left[x \sqrt{N} \psi_\ell^{(N)}(x) + \frac{1}{\sqrt{N}} \frac{d}{dx} \psi_\ell^{(N)}(x) \right] \\
\frac{d^2}{dy^2} H_\ell(y) &= c \frac{d}{dy} \left[\frac{d}{dy} \left[e^{y^2/2} \cdot \psi_\ell^{(N)}\left(\frac{y}{\sqrt{N}}\right) \right] \right] \\
&= c \frac{d}{dy} \left[e^{y^2/2} \left(y \psi_\ell^{(N)}\left(\frac{y}{\sqrt{N}}\right) + \frac{d}{dy} \psi_\ell^{(N)}\left(\frac{y}{\sqrt{N}}\right) \right) \right] \\
&= c \left[y e^{y^2/2} \left(y \psi_\ell^{(N)}\left(\frac{y}{\sqrt{N}}\right) + \frac{d}{dy} \psi_\ell^{(N)}\left(\frac{y}{\sqrt{N}}\right) \right) \right. \\
&\quad \left. + e^{y^2/2} \left(\psi_\ell^{(N)}\left(\frac{y}{\sqrt{N}}\right) + y \frac{d}{dy} \psi_\ell^{(N)}\left(\frac{y}{\sqrt{N}}\right) + \frac{d^2}{dy^2} \psi_\ell^{(N)}\left(\frac{y}{\sqrt{N}}\right) \right) \right] \\
&= c \left[(y^2 + 1) \psi_\ell^{(N)}\left(\frac{y}{\sqrt{N}}\right) + 2y \frac{d}{dy} \psi_\ell^{(N)}\left(\frac{y}{\sqrt{N}}\right) + \frac{d^2}{dy^2} \psi_\ell^{(N)}\left(\frac{y}{\sqrt{N}}\right) \right] \\
&= c \left[(x^2 N + 1) \psi_\ell^{(N)}(x) + 2x \frac{d}{dx} \psi_\ell^{(N)}(x) + \frac{1}{N} \frac{d^2}{dx^2} \psi_\ell^{(N)}(x) \right]
\end{aligned}$$

Then, plugging this into the ODE (removing common factor $c e^{y^2/2}$)

$$(x^2 N + 1) \psi_\ell^{(N)}(x) + 2x \cancel{\frac{d}{dx}} \psi_\ell^{(N)}(x) + \frac{1}{N} \frac{d^2}{dx^2} \psi_\ell^{(N)}(x) - 2x^2 N \psi_\ell^{(N)}(x) = 0$$

$$-2x \cancel{\frac{d}{dx}} \psi_\ell^{(N)}(x) + 2\ell \psi_\ell^{(N)}(x) = 0$$

$$\Rightarrow \frac{1}{N} \frac{d^2}{dx^2} \psi_\ell^{(N)}(x) + (x^2 N + 1 - 2x^2 N + 2\ell) \psi_\ell^{(N)}(x) = 0$$

$$\Rightarrow \frac{1}{N} \frac{d^2}{dx^2} \psi_\ell^{(N)}(x) = -(-x^2 N + (2\ell + 1)) \psi_\ell^{(N)}(x)$$

$$\therefore \frac{d^2}{dx^2} \psi_\ell^{(N)}(x) = (x^2 N - (2\ell + 1)) \psi_\ell^{(N)}(x) \quad \text{as required.}$$

3. //

Define

$$\psi_j(x) = N^{1/4} \gamma_j \exp\left(-\frac{N}{2} x^2\right) H_j(\sqrt{N} x)$$

$$\text{with } \gamma_j = \frac{1}{\sqrt{\pi} 2^j j!}.$$

Firstly, $\psi_{N-k}(x) = N^{1/4} \gamma_{N-k} e^{-Nx^2/2} H_{N-k}(\sqrt{N}x)$

$$= N^{1/4} \gamma_{N-k} e^{-N} \left(e^{-x^2/2} H_{N-k}(\sqrt{N}x) \right)$$

Define

$$t_k^{(N)}(x) := e^{-x^2/2} H_{N-k}(\sqrt{N}x)$$

Then use P-R asymptotic:

$$t_k^{(N)}(x) = 2^{\frac{N-k}{2} - \frac{1}{4}} \sqrt{(N-k)!} (\pi(N-k))^{-\frac{1}{4}} \\ \times \frac{1}{\sqrt{\sin \phi}} \left(\sin \left[\left(\frac{N-k}{2} + \frac{1}{4} \right) (\sin(2\phi) - 2\phi) + \frac{3\pi}{4} \right] + O((N-k)^{-1}) \right)$$

for $x = \sqrt{2(N-k)+1} \cos \phi$.

We also have

$$\gamma_{N-k} = \frac{1}{\sqrt{\pi} 2^{N-k} (N-k)!}$$

← cancels with the one in $t_k^{(N)}(x)$

Then

$$\psi_{N-k}(x) = N^{1/4} \pi^{-1/4} 2^{-\cancel{(N-k)/2}} e^{-N} \cdot 2^{\cancel{(N-k)/2}} 2^{-1/4} \pi^{-1/4} (N-k)^{-1/4} \frac{1}{\sqrt{\sin \phi}} \left(\sin \left[\left(\frac{N-k}{2} + \frac{1}{4} \right) (\sin 2\phi - 2\phi) + \frac{3\pi}{4} \right] + O((N-k)^{-1}) \right) \\ = \frac{e^{-N}}{\pi^{1/8} 2^{1/4}} \left(\frac{N}{N-k} \right)^{-1/4} \frac{1}{\sqrt{\sin \phi}} \left(\sin \left[\left(\frac{N-k}{2} + \frac{1}{4} \right) (\sin 2\phi - 2\phi) + \frac{3\pi}{4} \right] + O((N-k)^{-1}) \right)$$

Now consider limiting values as $N \rightarrow \infty$ of $\left(\frac{N}{N-k} \right)^{-1/4} \xrightarrow{N \rightarrow \infty} 1$ and $O((N-k)^{-1}) \xrightarrow{N \rightarrow \infty} O(N^{-1})$

$$\therefore \psi_{N-k}(x) = \frac{e^{-N}}{\pi^{1/8} 2^{1/4}} \frac{1}{\sqrt{\sin \phi}} \left(\sin \left[\left(\frac{N-k}{2} + \frac{1}{4} \right) (\sin 2\phi - 2\phi) + \frac{3\pi}{4} \right] + O\left(\frac{1}{N}\right) \right)$$

Now look at sin term in brackets.

$$\sin \left[\left(\frac{N-k}{2} + \frac{1}{4} \right) (\sin 2\phi - 2\phi) + \frac{3\pi}{4} \right]$$

$$= \sin \left[\frac{N-k}{2} \sin 2\phi - (N-k)\phi + \frac{1}{4} \sin 2\phi - \frac{1}{2}\phi + \frac{3\pi}{4} \right]$$

$\sin \alpha = \cos \left(\frac{\pi}{2} - \alpha \right)$, hence, the above becomes

$$\cos \left[\frac{\pi}{2} - \frac{N-k}{2} \sin 2\phi + (N-k)\phi - \frac{1}{4} \sin 2\phi + \frac{1}{2}\phi - \frac{3\pi}{4} \right]$$

$$= \cos \left[(N-k+\frac{1}{2})\phi - (\frac{N-k}{2} + \frac{1}{4})\sin 2\phi - \frac{\pi}{4} \right]$$

$$= \cos \left[N\left(\phi - \frac{1}{2}\sin 2\phi\right) + k\left(\frac{1}{2}\sin 2\phi - \phi\right) + \frac{1}{2}\left(\phi - \frac{1}{2}\sin 2\phi\right) - \frac{\pi}{4} \right]$$

$$= \cos \left[\left(N - k + \frac{1}{2}\right) \beta(\phi) - \frac{\pi}{4} \right]$$

↑ not sure what's happening here

Didn't really have time to finish this - not sure how the above becomes the form that we see in the result.